Ajithkumar P

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Professional Summary

Full Stack / MERN Stack Developer with **4 years of professional experience** building scalable web, mobile, SaaS, and eCommerce applications using **React.js, Node.js, Express.js, MongoDB, Redis, and Flutter**. Experienced in developing **Shopify applications from scratch**, including frontend and backend development, database schema design, Shopify APIs, App Blocks, theme customization, metafields, product-based pricing, and customer-specific customizations. Experienced in building modular backend services using **Microservices Architecture**, secure backend systems using **JWT authentication and Role-Based Access Control (RBAC)**, integrating third-party APIs, and implementing payment and subscription workflows with **PayPal and BigCommerce**.

Technical Skills

Languages: JavaScript, HTML5, CSS3, Dart, Liquid

Frontend: React.js, React Router, Flutter, Tailwind CSS

Backend: Node.js, Express.js, Sails.js, Microservices Architecture

Database & Caching: MongoDB, Redis

APIs: REST API, GraphQL, Shopify APIs, Third-Party APIs, Webhooks

Authentication & Security: JWT, RBAC, HTTP-only Cookies

E-Commerce & SaaS: Shopify, Shopify App Development, Shopify Polaris, Shopify Web Components, Shopify App Blocks, Theme Customization, BigCommerce

AI & Emerging Technologies: Google Gemini API (Multimodal), Model Context Protocol (MCP), MCP Server Development

Payments: PayPal, Razorpay

Cloud & Tools: Git, GitHub, Postman, Google Cloud Scheduler

Work Experience

Full Stack / MERN Stack Developer Dec 2022 – Present

Acumen Technologies Group Inc

- Developed and maintained scalable **SaaS and eCommerce applications** using React.js, Node.js, Express.js, and MongoDB.
- Designed and developed **Shopify applications from scratch**, covering frontend, backend, database schema design, business logic, and third-party integrations.
- Developed reusable React.js components and backend services using modular architecture and separation of concerns.
- Worked with **Microservices Architecture** to develop modular and maintainable backend services.
- Used **Redis** for caching and handling frequently accessed application data.
- Designed and implemented **RESTful APIs and GraphQL services** for application and eCommerce workflows.
- Integrated **Shopify APIs** for merchant, product, media, metafield, pricing, and application-specific workflows.
- Implemented **Shopify App Blocks and theme customization** to integrate application functionality into merchant storefronts.

- Developed **product-based pricing and billing workflows** based on merchant and business requirements.
- Designed and maintained **MongoDB database schemas** based on application requirements and business workflows.
- Integrated third-party APIs and webhooks to support payment, eCommerce, and business-specific workflows.
- Integrated **PayPal subscription functionality with BigCommerce** to support recurring monthly payment workflows.
- Automated Shopify product workflows including product creation, media uploads, and dynamic metafield management.
- Troubleshot production issues, implemented bug fixes, and delivered new features based on customer and business requirements.

Mobile Application Developer Aug 2022 – Dec 2022

Acumen Technologies Group Inc

- Developed and enhanced cross-platform mobile application features using **Flutter and Dart**.
- Implemented customer image-upload functionality with **free-form image cropping**, allowing customers to crop uploaded images according to their requirements.
- Developed image-processing functionality to integrate customer-uploaded images into the application workflow.
- Worked on **AR-based visualization functionality**, enabling customers to visualize images and frames on wall surfaces.
- Integrated mobile application features with existing backend APIs and collaborated with backend developers to ensure reliable data flow.

Professional Projects

LIVEARf - Art Framing – Shopify SaaS Application Shopify App

Tech Stack: React.js, Node.js, Express.js, MongoDB, Shopify APIs, Shopify App Blocks, Liquid, REST API

- Developed a **Shopify SaaS application from scratch** for artwork and image framing, enabling customers to upload or select artwork, choose frame and print options, and configure products before purchase.
- Built responsive frontend functionality using **React.js** and developed backend services using **Node.js and Express.js** to manage application workflows and customer configurations.
- Integrated **Shopify APIs** for product, merchant, media, metafield, and application-specific workflows, enabling communication between the application and Shopify storefronts.
- Implemented **Shopify App Blocks and theme customization** to embed application features directly into merchant storefronts.
- Designed and maintained **MongoDB database schemas** for images, artworks, frames, products, and customer configurations based on application and business requirements.
- Developed **product-based pricing and paid customization workflows**, supporting merchant-specific pricing requirements and additional one-time customization charges.
- Implemented **customer-specific features and storefront customizations** based on business requirements and external website references.
- Integrated **third-party APIs and webhooks** to support external services, application workflows, and business-specific integrations.
- Automated Shopify product workflows including **product creation, media uploads, and dynamic metafield management**.

- Implemented room-view functionality allowing customers to **visualize selected artwork and frames in different room and wall environments**.

LIVEARf Wall Gallery S+ – Shopify Application Shopify App

Tech Stack: React.js, Node.js, Express.js, MongoDB, Shopify APIs, Shopify App Blocks, Liquid, REST API

- Developed a **Shopify application for artwork and image upload, printing, framing, and wall-gallery visualization**, providing customers with an interactive product customization experience.
- Implemented **multiple image upload functionality**, allowing customers to select multiple artworks and create multi-image gallery wall configurations.
- Built interactive React.js functionality for **image positioning, gallery layouts, frame selection, and visualization**, enabling customers to configure gallery arrangements dynamically.
- Developed **Node.js and Express.js backend APIs** to manage uploaded images, artwork configurations, gallery layouts, frame selections, and application data.
- Integrated **Shopify APIs and App Blocks** to connect the application with merchant storefronts and provide customized gallery functionality.
- Designed and maintained **MongoDB schemas** for images, artworks, frames, gallery layouts, and customer configurations.
- Implemented reusable frontend and backend components to support **scalable gallery customization workflows** and simplify future feature enhancements.
- Implemented room and wall visualization functionality for **multi-image gallery arrangements**.
- Troubleshot application issues and delivered **production fixes and feature enhancements** based on merchant and business requirements.

Personal Projects

User Management MCP Server GitHub

Tech Stack: Node.js, Express.js, Model Context Protocol (MCP), MCP SDK, Zod, Streamable HTTP

- Built a **User Management MCP Server** using Node.js, Express.js, and the Model Context Protocol (MCP) SDK to expose user management capabilities to AI clients.
- Developed MCP **Tools** for creating, retrieving, updating, and deleting users, enabling AI clients to perform user management operations through structured tool calls.
- Implemented **Zod-based input validation** to validate tool parameters and enforce structured data requirements before executing backend operations.
- Implemented an MCP **Resource** containing a user management API guide with available tools, required parameters, and usage instructions for AI clients.
- Implemented an MCP **Prompt** for generating structured user creation requests and providing predefined instructions to AI clients.
- Implemented **Streamable HTTP transport** to enable communication between MCP clients and the MCP server through an HTTP endpoint.
- Integrated the MCP server with **Cursor** and tested MCP tool discovery, tool invocation, resource access, prompt execution, and backend workflows.
- Structured the application into reusable backend components to support **maintainable MCP tool, resource, and prompt implementations**.

MERN Ecommerce Platform Live Demo GitHub

Tech Stack: React.js, Node.js, Express.js, MongoDB, Redux Toolkit, JWT, Razorpay, REST API

- Developed a full-stack **MERN eCommerce application** using React.js, Node.js, Express.js, and MongoDB with responsive user and administrative workflows.
- Implemented **JWT authentication and Role-Based Access Control (RBAC)** to secure user accounts, protected routes, and role-specific application functionality.
- Developed complete eCommerce workflows including **product catalog, product search, shopping cart, checkout, and order management**.
- Built an **admin dashboard** for managing products, users, categories, and customer orders through centralized administrative workflows.
- Integrated **Razorpay payment gateway** to support secure online payment processing during checkout.
- Implemented reusable React.js components and application routing to maintain a **structured and scalable frontend architecture**.
- Integrated frontend and backend services through **REST APIs** for authentication, product management, cart, checkout, and order workflows.

AI Image SEO Generator Live Demo GitHub

Tech Stack: React.js, Node.js, Express.js, Google Gemini API, Multer, REST API

- Developed an **AI-powered image SEO application** that analyzes uploaded images and automatically generates SEO metadata for digital content.
- Integrated **Google Gemini Multimodal AI** to analyze image content and generate structured SEO information including titles, descriptions, tags, and categories.
- Implemented image upload processing using **Multer with in-memory buffer handling**, allowing uploaded image data to be processed by the backend before AI analysis.
- Developed a modular Node.js and Express.js backend using **routes, controllers, and service layers** to separate API, business logic, and AI integration responsibilities.
- Implemented REST API endpoints for **image upload and AI-generated metadata processing** between the frontend and backend.
- Structured the Gemini integration as a dedicated service to maintain **reusable and maintainable AI processing logic**.

JWT Authentication API GitHub

Tech Stack: Node.js, Express.js, MongoDB, JWT, bcrypt, HTTP-only Cookies

- Developed a standalone **RESTful authentication API** using Node.js, Express.js, MongoDB, and JWT for secure user authentication and authorization.
- Implemented **secure password hashing using bcrypt** to protect stored user credentials.
- Implemented **JWT-based access and refresh token authentication** to maintain authenticated user sessions.
- Used **HTTP-only cookies** for secure refresh-token storage and reduced exposure of authentication tokens to client-side scripts.
- Developed authentication and authorization middleware to protect **REST API routes and enforce user permissions**.
- Designed reusable authentication components for **login, authorization, protected routes, and token validation** workflows.

Education

B.E. Electronics and Communication Engineering 2018 – 2022

University V.O.C College of Engineering CGPA: 7.22/10